`CausalGym`: Benchmarking causal interpretability methods on linguistic tasks

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Abstract

Language models (LMs) have proven to be powerful tools for psycholinguistic research, but most prior work has focused on purely behavioural measures (e.g., surprisal comparisons). At the same time, research in model interpretability has begun to illuminate the abstract causal mechanisms shaping LM behavior. To help bring these strands of research closer together, we introduce **`CausalGym`**. We adapt and expand the SyntaxGym suite of tasks to benchmark the ability of interpretability methods to causally affect model behaviour. To illustrate how **`CausalGym`** can be used, we study the `pythia` models (14M–6.9B) and assess the causal efficacy of a wide range of interpretability methods, including linear probing and distributed alignment search (DAS). We find that DAS outperforms the other methods, and so we use it to study the learning trajectory of two difficult linguistic phenomena in `pythia-1b`: negative polarity item licensing and filler–gap dependencies. Our analysis shows that the mechanism implementing both of these tasks is learned in discrete stages, not gradually.

`https://github.com/aryamanarora/causalgym`

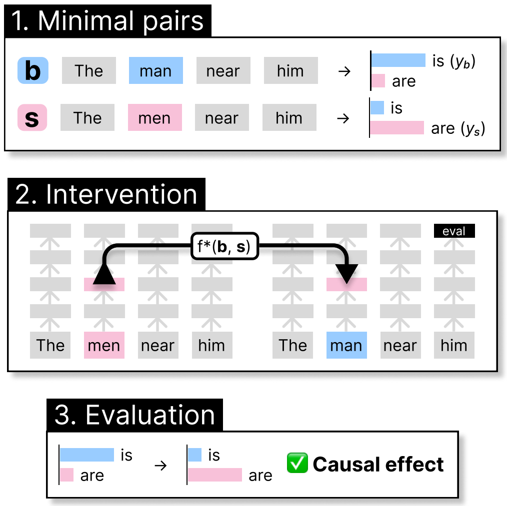

Figure 1: The **`CausalGym`** pipeline: **(1)** take an input minimal pair $(\mathbf{b}, \mathbf{s})$ exhibiting a linguistic alternation that affects next-token predictions (y_b, y_s); **(2)** intervene on the base forward pass using a pre-defined intervention function that operates on aligned representations from both inputs; **(3)** check how this intervention affected the next-token prediction probabilities. In aggregate, such interventions assess the causal role of the intervened representation on the model's behaviour.

1 Introduction

Language models have found increasing use as tools for psycholinguistic investigation—to model word surprisal (Smith and Levy, 2013; Goodkind and Bicknell, 2018; Wilcox et al., 2023a; Shain et al., 2024, *inter alia*), graded grammaticality judgements (Hu et al., 2024), and, broadly, human language processing (Futrell et al., 2019; Warstadt and Bowman, 2022; Wilcox et al., 2023b). To benchmark the linguistic competence of LMs, computational psycholinguists have created **targeted syntactic evaluation** benchmarks, which feature minimally-different pairs of sentences differing in grammaticality; success is measured by whether LMs assign higher probability to the grammatical sentence in each pair (Marvin and Linzen, 2018). Despite the increasing use of LMs as models of human linguistic competence and how much easier it is to experiment on them than human brains, we do not understand the mechanisms underlying model behaviour—LMs remain largely uninterpretable.

The **linear representation hypothesis** claims that 'concepts' form linear subspaces in the representations of neural models. An increasing body of experimental evidence from models trained on language and other tasks supports this idea (Mikolov et al., 2013; Elhage et al., 2022; Park et al., 2023; Nanda et al., 2023). Per this hypothesis, information about high-level linguistic alternations should be localised to linear subspaces of LM activations.

Methods for finding such features, and even modifying activations in feature subspaces to causally influence model behaviour, have proliferated, including probing (Ettinger et al., 2016; Adi et al., 2017), distributed alignment search (DAS; Geiger et al., 2023b), and difference-in-means (Marks and Tegmark, 2023).

Psycholinguistics and interpretability have complementary needs: thus far, psycholinguists have evaluated LMs on extensive benchmarks but neglected understanding their internal mechanisms, while new interpretability methods have only been evaluated on one-off datasets and so need better benchmarking. Thus, we introduce **`CausalGym`** (Figure 1). We adapt linguistic tasks from SyntaxGym (Gauthier et al., 2020) to benchmark interpretability methods on their ability to find linear features in LMs that, when subject to intervention, causally influence linguistic behaviours. We study the `pythia` family of models (Biderman et al., 2023), finding that DAS is the most efficacious method. However, our investigation corroborates prior findings that DAS is powerful enough to make the model produce arbitrary input–output mappings (Wu et al., 2023). To address this, we adapt the notion of control tasks from the probing literature (Hewitt and Liang, 2019), finding that adjusting for performance on the arbitrary mapping task reduces the gap between DAS and other methods.

We further investigate how LMs learn two difficult linguistic behaviours during training: filler–gap extraction and negative polarity item licensing. We find that the causal mechanisms require multi-step movement of information, and that they emerge in discrete stages (not gradually) early in training.

2 Related work

Targeted syntactic evaluation. Benchmarks adhering to this paradigm include SyntaxGym (Gauthier et al., 2020; Hu et al., 2020), BLiMP (Warstadt et al., 2020), and several earlier works (Linzen et al., 2016; Gulordava et al., 2018; Marvin and Linzen, 2018; Futrell et al., 2019). We use the SyntaxGym evaluation sets over BLiMP even though the latter has many more examples, because we require minimal pairs that are grammatical sentences alternating along a specific feature (e.g. number). Such pairs be templatically constructed with minimal adaptation using SyntaxGym's format.

Interventional interpretability. Interventions are the workhorse of causal inference (Pearl, 2009), and have thus been adopted by recent work in interpretability for establishing the causal role of neural network components in implementing certain behaviours (Vig et al., 2020; Geiger et al., 2021, 2022, 2023a; Meng et al., 2022; Chan et al., 2022; Goldowsky-Dill et al., 2023), particularly linguistic ones like coreference and gender bias (Lasri et al., 2022; Wang et al., 2023; Hanna et al., 2023; Chintam et al., 2023; Yamakoshi et al., 2023; Hao and Linzen, 2023; Chen et al., 2023; Amini et al., 2023; Guerner et al., 2023). The approach loosely falls under the nascent field of *mechanistic interpretability*, which seeks to find interpretable mechanisms inside neural networks (Olah, 2022).

We illustrate the interventional paradigm in Figure 1; given a base input $\mathbf{b}$ and source input $\mathbf{s}$, all interventional approaches take a model-internal component f and replace its output with that of $f^*(\mathbf{b}, \mathbf{s})$, which modifies the representation of $\mathbf{b}$ using that of $\mathbf{s}$. The core idea of intervention is adopted directly from the do-operator used in causal inference; we test the intervention's effect on model output to establish a causal relationship.

3 Benchmark

To create **`CausalGym`**, we converted the core test suites in SyntaxGym (Gauthier et al., 2020) into templates for generating large numbers of span-aligned minimal pairs, a process we describe below along with our evaluation setup.

3.1 Premise

Each test suite in SyntaxGym focuses on a single linguistic feature, constructing English-language minimal pairs that minimally adjust that feature to change expectations about how a sentence should continue. A test suite contains several *items* which share identical settings for irrelevant features, and each item has some *conditions* which vary only the important feature. All items adhere to the same templatic structure, sharing the same ordering and set of *regions* (syntactic units). To measure whether models match human expectations, SyntaxGym evaluates the model's surprisal at specific regions between differing conditions.

For example, the *Subject-Verb Number Agreement (with prepositional phrase)* task constructs items consisting of 4 conditions, which set all possible combinations of the number feature on subjects

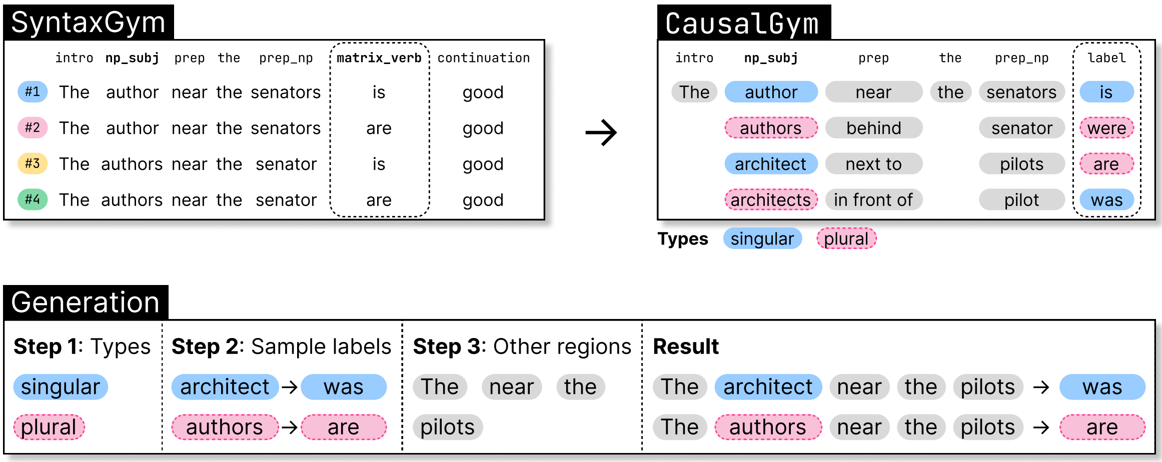

Figure 2: An example of the **CausalGym** conversion process on the test suite *Subject-Verb Number Agreement (with prepositional phrase)*. The left side shows how items are structured in SyntaxGym originally, which we process into the templatic format on the right. The bottom shows how we sample a minimal pair.

and their associated verbs, as well as the *opposite* feature on a distractor noun. Each example in this test suite follows the template

(1) The `np_subj` `prep` the `prep_np` `matrix_verb` `continuation`.

where, in a single item, the regions `np_subj` and `matrix_verb` are modified along the number feature, and `prep_np` is a distractor. For example:

(2) The **author** near the senators **is** good.

(3) *The **author** near the senators **are** good.

(4) *The **authors** near the senator **is** good.

(5) The **authors** near the senator **are** good.

Humans expect agreement between the number feature on the verb and the subject, as in (2) and (5). On this test suite, SyntaxGym measures if the surprisal at the verb satisfies the following inequalities between conditions: $p(\text{is} \mid \text{author}) > p(\text{are} \mid \text{author})$ and $p(\text{are} \mid \text{authors}) > p(\text{is} \mid \text{authors})$.

3.2 Templatising SyntaxGym

Our goal is to study how LMs implement mechanisms for converting feature alternations in the input into corresponding alternations in the output—e.g., how does an LM keep track of the number feature on the subject when it needs to output an agreeing verb? In adapting SyntaxGym for this purpose, we need to address two issues: (1) to study model mechanisms, we only want *grammatical* pairs of sentences; and (2) SyntaxGym test suites contain < 50 items, while we need many more for training supervised interpretability methods and creating non-overlapping test sets.

Thus, we select the two grammatical conditions from each item and simplify the behaviour of interest into an explicit input–output mapping. For example, we recast *Subject-Verb Number Agreement (with prepositional phrase)* into counterfactual pairs that elicit singular or plural verbs based on the number feature of the subject, and hold everything else (including the distractor) constant:

(6) a. The **author** near the senators ⇒ **is**
b. The **authors** near the senators ⇒ **are**

To be able to generate many examples for training, we use the aligned regions as slots in a template that we can mix-and-match between items to combinatorially generate pairs, illustrated in Figure 2. We manually removed options that would have resulted in questionably grammatical sentences.

For generation using our format, each template has a set of types T which govern the input **label variable** and the expected next-token prediction **label**. To generate a counterfactual pair, we first sample two types $t_1, t_2 \sim T$ such that $t_1 \neq t_2$. Then, for the label variable and label, we sample an option of that type t_1 (for the first sentence) or t_2 (for the second). Finally, for the non-label variable regions, we sample one option and set both sentences to that. In Figure 2, we show the generation process in the bottom panel; types for the label variable and label options are colour-coded.